Subject: DATABASE MANAGEMENT SYSTEMS
Type: Exam Preparation Summary
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Unit 3: Normalization & Indexing Summary

1. Functional Dependencies (FDs)

An FD X -> Y holds on relation R if whenever two tuples agree on attribute X, they must also agree on attribute Y. Armstrong's Axioms: Reflexivity, Augmentation, Transitivity. Secondary rules: Decomposition, Union, Pseudotransitivity.

2. Normal Forms Hierarchy

1NF: Atomic values only, no repeating groups.

2NF: In 1NF and no partial dependencies (every non-prime attribute is fully functionally dependent on every candidate key).

3NF: In 2NF and no transitive dependencies (for every FD X -> A, either X is a superkey or A is prime).

BCNF: For every non-trivial functional dependency X -> A, X must be a superkey.

3. Indexing & B+ Trees

Primary vs Secondary Index, Dense vs Sparse Index. B+ Trees maintain balance with height O(log N) supporting logarithmic search, insertion, and range queries.

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